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On modelling ballistic gelatine by smoothed particle hydrodynamics (SPH)

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Introduction

- Motivation

Smoothed particle hydrodynamics

- Basic theory of SPH

- Viscous and incompressible matter

- High-speed loading

- Modelling ballistic gelatine

- Numerical model setup

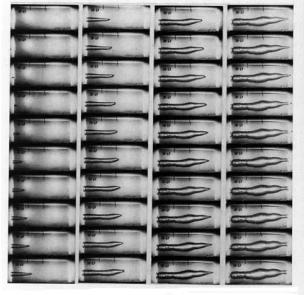
Results

- Comparison to experimental data

Conclusion

- Acknowledgements

- ▶ Penetrating ballistic impact for estimating consequences and severity of gunshot wounds
- ▶ Ethics motivates for using substituting matter and numerical modelling
- ▶ Ballistic gelatine correlated well to human muscle tissue
- ▶ Smoothed particle hydrodynamics
 - ▶ Ballistic gelatine highly sensitive to strain rate and temperature
 - ▶ Behaving as shear thickening fluid at high shear rates
 - ▶ Large deformations and material discontinuities
- ▶ Numerical modelling of ballistic gelatine for future use in virtual testing



- ▶ Mainly for astrophysical problems
- ▶ Highly deformable solids
- ▶ Considerably moving matter and boundaries
- ▶ Conservation laws transformed into integral equations by kernel function $w(\mathbf{r}, h)$ with $\lim_{h \rightarrow 0} w(\mathbf{r}, h) = \delta(\mathbf{r})$
- ▶ $w \sim$ weighting function defining how much of each field variable contributes to field variable at given point in space

$$\langle A(\mathbf{r}) \rangle = \int_V A(\mathbf{r}') w(\mathbf{r} - \mathbf{r}', h) d\mathbf{r}'$$

$$\langle A(\mathbf{r}) \rangle = \sum_{j=1}^N \frac{m_j}{\rho_j(\mathbf{r}_j)} A(\mathbf{r}_j) w(\mathbf{r} - \mathbf{r}_j, h)$$

$$\langle \nabla_r A(\mathbf{r}) \rangle = \sum_{j=1}^N \frac{m_j}{\rho_j(\mathbf{r}_j)} A(\mathbf{r}_j) \nabla_r w(\mathbf{r} - \mathbf{r}_j, h)$$

- ▶ Equation of motion $\frac{d\mathbf{v}_i}{dt} = - \sum_{j=1}^N m_j \left(\frac{p_j}{\rho_j^2} + \frac{p_i}{\rho_i^2} + \Pi_{ij} \right) \nabla_r w_{ij}$
- ▶ Continuity equation $\frac{d\rho_i}{dt} = \sum_{j=1}^N m_j (\mathbf{v}_i - \mathbf{v}_j) \nabla_r w_{ij}$
- ▶ Equation of conserving energy $\frac{du_i}{dt} = \sum_{j=1}^N m_j \frac{p_j}{\rho_j^2} (\mathbf{v}_i - \mathbf{v}_j) \nabla_r w_{ij}$
- ▶ Artificial viscosity $\Pi_{ij} = \frac{1}{\rho_{ij}(-\alpha\mu_{ij}c_{ij} + \beta\mu_{ij}^2)}$, $A_{ij} = \frac{A_i - A_j}{2}$
- ▶ Finding nearest particles contributing to physical value of particular particle
- ▶ Optimised nearest neighbour search



- ▶ Second derivative challenging
- ▶ Run in quasi-incompressible limit
$$p = K \left[\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right]$$
$$K = \frac{c^2 \rho_0}{\gamma}, \gamma = 7$$
- ▶ K takes into account density fluctuations under 1% for weakly compressible fluid
- ▶ For maximum flow speed $v \frac{\delta p}{\rho_0} = \frac{v^2}{c^2} = M^2$

- Polynomial Mie-Grüneisen equation of state

$$p = \begin{cases} \sum_{k=0}^3 C_k \eta^k + (C_4 + C_5 \eta) E_n & \forall \eta \geq 0 \\ C_0 + C_1 \eta & \forall \eta < 0 \end{cases}$$
$$\eta = \frac{\rho}{\rho_0} - 1$$



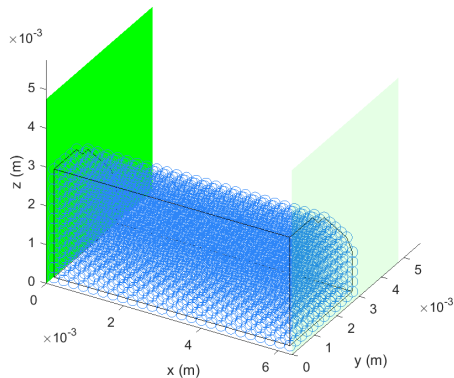
- ▶ Hydrodynamic material
- ▶ Highly viscous matter $\mu \sim 10 \text{ m} \cdot \text{Pa} \cdot \text{s}$
for quasi-static loading
- ▶ Dynamic viscosity $\mu \sim 1000\times$ higher
- ▶ Small cylinder sample
 $h = d = 6.35 \text{ mm}$

Numerical model setup

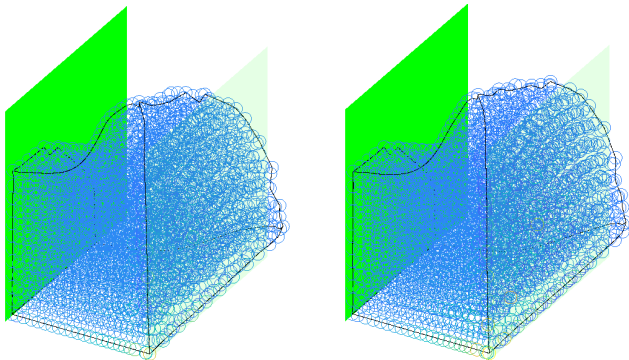


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- ▶ Symmetry in 2 planes
- ▶ Uniform particle spacing
 $\Delta = 0.254 \text{ mm}$
- ▶ 3025 particles in 3D
- ▶ Frictionless contact between domain and plates
- ▶ Pre-locating neighbouring particles
- ▶ Static $\dot{\epsilon} = 0.0 \text{ s}^{-1}$
- ▶ Dynamic $\dot{\epsilon} = 1000 \text{ s}^{-1}$



- Agreement to experimental data until $\epsilon = 0.5$ for both quasi-static and dynamic loadings



- Obvious non-uniform deformation

- ▶ SPH implemented for ballistic gelatine numerical model
- ▶ Generic material properties of ballistic gelatine
- ▶ Numerical simulations verified the material model
 - ▶ Quasi-static compression
 - ▶ Dynamic compressions

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